INCH-POUND

MIL-DTL-14506H
18 February 1997
SUPERSEDING
MIL-H-0014506G(EA)
19 September 1991
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8 January 1970

DETAIL SPECIFICATION

HOSE, AIR DUCT: FOR FILTER UNITS

Inactive for new design after 11 March 1996

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a type of low pressure flexible hose fabricated in specific lengths.

1.2 Classification. The hose shall be made in the following lengths as specified (see 6.2):

Size 1 – 2 feet long
Size 2 – 6 feet long
Size 3 – 9 feet long
Size 4 – 4 feet long
Size 5 – 3-3/4 inches long
Size 6 – 6 inches long

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to insure the completeness of this list, document users are cautioned that they must meet

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Technical Director, U.S. Army Edgewood Research, Development and Engineering Center, ATTN: SCBRD-ENE-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4240

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all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.1.1 Government documents, drawings, and publications. The following Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. ARMY EDGEWOOD RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

DRAWINGS

5-19-916	- Hose - Assembly and Detail
136-41-3000	- Q170, Apparatus, Static Diffusion

(Copies are available from Technical Director, U.S. Army Edgewood Research, Development and Engineering Center, ATTN: SCBRD-ENE-M, Aberdeen Proving Ground, MD 21010-5423.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials and components.

3.2.1 Materials. All materials cited on Drawing 5-19-916 shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawing.

3.2.1.1 Outer covering. The outer covering shall be tubular stockinet made from nylon. The nylon stockinet shall be dark green (see 6.3) and meet the following physical requirements:

Denier	210
Weight per square yard	4.4 ounces minimum

3.2.2 Components. All components of the hose shall conform to the specifications and drawing listed on Drawing 5-19-916.

3.3 Manufacture and assembly. The hose shall be manufactured and assembled as specified on Drawing 5-19-916.

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3.4 Physical properties.

3.4.1 Weight. The weight of the hose shall not exceed 5.0 ounces per foot of the original unstressed length when tested as specified in 4.3.6.1.

3.4.2 Resilience. The permanent set of the hose shall not exceed 4 percent of the original unstressed length when tested as specified in 4.3.6.2.

3.4.3 Extensibility. The length of the hose shall increase at least 36 percent of the original unstressed length when tested as specified in 4.3.6.3.

3.4.4 Crush–resistance and permanent distortion. The hose shall maintain a minimum of 90 percent of its unstressed diameter and return to within 98 percent of its unstressed diameter within 2 minutes after load is removed when tested as specified in 4.3.6.4.

3.4.5 Flexibility. The hose shall be flexible when tested as specified in 4.3.6.5. This requirement is not applicable to the size 5 or size 6 hose.

3.4.6 Low–temperature fracture resistance. The hose shall not fracture when tested as specified in 4.3.6.6.

3.4.7 Leakage. The hose shall not leak when subjected to internal pressure of 5 ± 1/2 pound per square inch gage (psig) for a period of 3 minutes after stabilization when tested as specified in 4.3.6.7.

3.4.8 Freedom from odor. The hose shall be free from objectionable odor when tested as specified in 4.3.6.8.

3.4.9 Mustard resistance. The mustard–vapor resistance of the hose shall be at least 30 minutes when tested as specified in 4.3.6.9.

3.5 Workmanship. The hose shall be free from incomplete or defective vulcanizing, damage such as cut, torn, worn or abraded stockinet, cuts or cracks or separation in the rubber end covering and foreign matter such as dirt, oil, or viscous material inside or outside the hose.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2)
(b) Conformance inspection (see 4.3)

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4.2 First article inspection.

4.2.1 Inspections to be performed. As determined by the Government, the first article sample items shall be examined and tested in accordance with this specification using either the first article procedures in 4.2.3 or other first article procedures which will be provided by the Government. As determined by the Government, the sample first article items may be subjected to any or all of the examinations and tests specified in this specification and be inspected for compliance with any or all of the requirements of the applicable drawings.

4.2.2 Sample. The first article sample shall consist of 20 hoses of the size or sizes specified and shall be manufactured using the same methods, materials, equipment, and processes as will be used during regular production (see 6.2). The first article sample shall be submitted for inspection and approval in accordance with the terms of the contract.

4.2.3 Inspection procedures. Each hose in the first article sample shall be examined for compliance with the workmanship requirement (3.5) of this specification and all the requirements of the applicable drawings. Hoses, selected at random from the first article sample, shall be subjected to the tests specified in table I.

TABLE I. First article inspection procedures

Inspection procedure	Number of first article items
Physical properties (4.3.6.1 – 4.3.6.8)	5 hoses
Mustard resistance (4.3.6.9 and 6.4)	5 hoses

4.2.4 Acceptance criteria. If any first article sample item fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure to comply with any of the requirements. The contractor shall obtain written approval from the contracting activity prior to proceeding with regular production.

4.3 Conformance inspection.

4.3.1 Lotting. A lot shall consist of the items produced by one manufacturer, at one plant, from the same materials, and under the same manufacturing conditions.

4.3.2 Sampling. Sampling shall be conducted in accordance with the classification of characteristics in 4.3.5 and, when specified, table II. Samples shall be selected at random.

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TABLE II. Sampling

Lot size	Inspection levels and sample sizes										
	I	II	III	IV	V	VI	VII	VIII	IX	X	XI
2 to 8	*	*	*	*	*	*	*	*	5	3	2
9 to 15	*	*	*	*	*	*	13	8	5	3	2
16 to 25	*	*	*	*	*	20	13	8	5	3	3
26 to 50	*	*	*	*	32	20	13	8	5	5	5
51 to 90	*	*	*	50	32	20	13	8	7	6	5
91 to 150	*	*	125	50	32	20	13	12	11	7	6
151 to 280	*	*	125	50	32	20	20	19	13	10	7
281 to 500	*	315	125	50	48	47	29	21	16	11	9
501 to 1200	*	315	125	75	73	47	34	27	19	15	11
1201 to 3200	1250	315	125	116	73	53	42	35	23	18	13
3201 to 10000	1250	315	192	116	86	68	50	38	29	22	15
10001 to 35000	1250	315	294	135	108	77	60	46	35	29	15
35001 to 150000	1250	490	294	170	123	96	74	56	40	29	15
150001 to 500000	1250	715	345	200	156	119	90	64	40	29	15
500001 and over	1250	715	435	244	189	143	102	64	40	29	15

*Indicates one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection.
Accept the lot represented on zero nonconforming characteristics and reject the lot represented on one or more nonconforming characteristics for all inspection levels.

4.3.3 Inspection procedure. Every item in the lot shall be inspected for critical characteristics. The lot represented shall be rejected when nonconformance to a critical characteristic is found. Sample items and, when applicable, the packaging thereof shall be examined and tested in accordance with the classification of characteristics in 4.3.5. Failure of any sample item to conform to any characteristic in the classification of characteristics based on the sampling and acceptance criteria specified therein shall be cause for rejection of the lot represented.

4.3.4 Inspection characteristics. Critical characteristics are characteristics whose nonconformance to specified requirements is likely to result in hazardous or unsafe conditions for individuals using, maintaining, or depending upon the product or whose nonconformance to specified requirements is likely to prevent performance of the tactical function of a major end item. Major characteristics are characteristics whose nonconformance to specified requirements is likely to result in failure or to reduce materially the usability of the item for its intended purpose. Minor characteristics are characteristics whose nonconformance to specified

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requirements is not likely to reduce materially the operation or usability of the item for its intended purpose.

4.3.5 Classification of characteristics. Conformance examinations and tests shall be as specified in the following classification of characteristics paragraphs. When specified herein, accept on 0 and reject on 1 attributes sampling inspection shall be performed on the designated characteristics using the stated levels in table II for selection of sample sizes.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 5–19–916
4.3.5(a)	Hose		NEXT HIGHER ASSY

CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	None defined			
Major				
101	Specified components correctly assembled	Table II, Level VII	3.2, 3.3	VI
102	Inside diameter at ends	Table II, Level VII	3.3	CE
103	Length of ends	Table II, Level VII	3.3	CE
104	Overall length	Table II, Level VII	3.3	CE
105	Workmanship	Table II, Level VII	3.5	VI
106	Weight	Table II, Level IX	3.4.1	4.3.6.1
107	Resilience	Table II, Level IX	3.4.2	4.3.6.2
108	Extensibility	Table II, Level IX	3.4.3	4.3.6.3
109	Crush resistance and permanent distortion	Table II, Level IX	3.4.4	4.3.6.4
110	Flexibility	Table II, Level IX	3.4.5	4.3.6.5**
111	Low–temperature fracture resistance	Table II, Level IX	3.4.6	4.3.6.6
112	Leakage	Table II, Level IX	3.4.7	4.3.6.7
113	Freedom from odor	Table II, Level IX	3.4.8	4.3.6.8
114	Mustard resistance	2*; Acc 0, rej 1	3.4.9	4.3.6.9***
Minor				
201	Outside diameter at ends	Table II, Level IX	3.3	CE
202	Maximum outside diameter	Table II, Level IX	3.3	CE

NOTES:
*Sample hoses shall be taken at random form the hoses that have passed the tests specified in 4.3.6.1 through 4.3.6.8, inclusive.
**Not applicable to size 5 or size 6 hose. ***Destructive test.
CE – Commercial inspection equipment
VI – Visual inspection

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4.3.6 Tests. Tests shall be conducted in the sequence listed:

4.3.6.1 Weight. The weight of the hose shall be as specified when the hose is weighed to the nearest ounce.

4.3.6.2 Resilience. The hose shall be suspended by one end, and a weight of 5 pounds shall be attached to the other end and permitted to hang freely. After 100 ± 20 seconds, the weight shall be detached and the hose shall be left at rest for 100 ± 20 seconds. The length of the hose shall then be determined.

4.3.6.3 Extensibility. The hose shall be suspended by one end and a weight of 10 pounds shall be attached to the other end and allowed to hang freely. After 100 ± 20 seconds, the length of the hose shall be determined.

4.3.6.4 Crush–resistance and permanent distortion. A 4–inch length of the wire reinforced portion of hose shall be subjected to a 225 pound static load uniformly applied for 2 minutes. During this time the stressed outer diameter shall be compared to the unstressed outer diameter to determine crush resistance. Two minutes after the load is removed the stressed outer diameter shall be compared to the unstressed outer diameter to determine permanent distortion. The size 5 hose shall be subjected to a 100 pound static load uniformly applied over the wire reinforced section of the hose to determine crush resistance and permanent distortion.

4.3.6.5 Flexibility. (This test is not applicable to the size 5 or size 6 hose.)

(a) **Rotation.** In a room maintained at 25 ± 5°C, the hose shall be placed on a horizontal, flat surface with 20 inches of its length overhanging the edge. The part of the hose on the flat surface shall be rotated through 360 degrees about its longitudinal axis, at a uniform rate of approximately 3 revolutions per minute (rpm). The hose shall be rotated a minimum of six revolutions. At no time during the rotation shall the lower half of the overhanging portion deviate from the vertical by more than 10 degrees.

(b) **Coiling.** In a room maintained at 25 ± 5°C, the hose shall be closely coiled around a rod 2 inches in diameter. There shall be neither any exposure of the wire in the hose nor any other damage. After release, there shall be no noticeable permanent set in the hose.

4.3.6.6 Low–temperature fracture resistance. The hose shall be subjected to a temperature of −55 ± 5°C for 48 hours. Maintained at the aforementioned temperature, at least 20 inches of the hose of size 1 through size 4 shall be closely coiled around a rod 2 inches in diameter. The size 5 or size 6 hose shall be fixed, by hand, to simulate coiling around a 2 inch diameter rod. The hose shall be allowed to return to room temperature and then tested for leakage as specified in 4.3.6.7. Failure of the leakage test shall be regarded as failing the low temperature requirement.

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4.3.6.7 Leakage. The hose shall be tested as follows: Plug one end of the hose. Connect the other end to a controlled source of air pressure with a shut−off valve and a pressure gage. Make sure all connections are tight. Subject the hose to the required pressure. Close the shut−off valve. Allow a minimum of 2 minutes for the air in the hose to reach equilibrium. There may be an initial pressure drop during the equalization period. Observe the pressure gage for a minimum period of 3 minutes; any pressure drop shall indicate leakage.

4.3.6.8 Freedom from odor. The hose shall be free from objectionable odors when determined by smelling.

4.3.6.9 Mustard resistance. The mustard vapor resistance of the hose shall be determined by the Government as follows (see 6.4):

(a) **Preparation of S−328 solution.** Mix 0.5 gm of S−328 (tetrachloro−7, 8−diphenylglycoluril) and approximately 0.1 gm of anhydrous sodium or potassium carbonate with 10 mL of 1,1,2,2−tetrachloroethane. Shake the mixture thoroughly and filter through a No. 42 filter paper. Pour the clear filtrate into a small actinic glass bottle. The bottle should be kept closed at all times when not in use. A fresh solution should be prepared weekly.

(b) **Procedure.** Prepare strips of congo red indicator paper by dipping 4−inch (± 1/8−inch) by 1−1/2−inch (± 1/8−inch) strips of qualitative analytical filter paper for coarse, gelatinous or medium precipitates into approximately 0.1 percent solution of congo red in distilled water. When the paper is thoroughly impregnated, allow to air dry. The treated paper may be kept indefinitely but must be prepared and stored in an acid−free and mustard−free atmosphere. For all samples, except for the 6−inch long hose which shall be tested without cutting, cut a 6−inch (± 1/4−inch) long specimen from the reinforced portion of the hose between the soft ends, retaining the nylon stocking covering. Stopper one end of the specimen and condition the specimen and mustard to 25.0 ± 0.5 °C. Just before performing the test, dip a cotton swab into S−328 solution (see (a) above) and place a row of five spots on the indicator paper along the long axis of the strip in the middle 2 inches of its length. Allow the S−328 to dry, and with two rubber bands, attach the spotted strip of indicator paper to the hose specimen with its long axis parallel to and centered on the length of the hose specimen. The side of the paper to which the S−328 was applied shall be away from the hose. Place the hose specimen with attached indicator paper on a glass plate with the indicator paper against the glass and block in place with two rubber stoppers. With a Pasteur pipet, take 1/2 cubic centimeter (cc) of liquid mustard, distilled to not less than 95 percent purity, and distribute it along the middle 2 inches of the bottom interior surface of the hose specimen. Stopper the open end of the specimen and place the glass plate into the Q170 tester (Drawing 136−41−3000) which is thermostated to 37.0 ± 0.5 °C. Observe the spotted portion of the indicator paper strip for the development of a blue color in the spotted areas which indicates mustard permeation. Observe the test to permeation or for a period equal to the minimum requirement plus 10 minutes.

CAUTION: Mustard produces severe blister action in both the liquid and vapor phases. All

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mustard containers should be kept in a well ventilated hood when not in use. Great care should be exercised to prevent either the liquid mustard or its vapor from coming in contact with the body. Personnel should wear butyl rubber gloves in all operations involving the handling of mustard. The glove protocol in effect for the standing operating procedures being used should be strictly observed.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD–ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. These hoses are used in Nuclear Biological Chemical protective systems as flexible, agent–resistant air ducts that provide adequate flow rates of purified air to one individual.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced.
(c) Size of hose required.
(d) First article:

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract when testing is performed by the Government.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production.

(e) Packaging requirements (see 5.1).

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6.3 Color. The color of the nylon stockinet should approximate color chip 34079 of FED–STD–595 – Colors.

6.4 Mustard resistance. Sample hoses for agent resistance testing should be forwarded to Technical Director, U.S. Army Edgewood Research, Development and Engineering Center, ATTN: SCBRD–ENM–S, Building E5100, Aberdeen Proving Ground, MD 21010–5423.

6.5 International interest. Certain provisions of this specification are the subject of international standardization agreement QSTAG–216. When amendment, revision, or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels, including departmental standardization offices, to change the agreement of make other appropriate accommodations.

6.6 Subject term (key word) listing.

agent–resistant
chemical–biological
flexible
mask
mustard gas

6.7 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:

Army – EA
Navy – SH
Air Force – 99

Preparing activity:

Army – EA

(Project 4240–0651)

Review Activities:

Navy – MC
Air Force – 84
DLA – CS, GS

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INSTRUCTIONS

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-DLT-14506H	2. DOCUMENT DATE (YYMMDD) 970218

3. DOCUMENT TITLE HOSE, AIR DUCT: FOR FILTER UNITS

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
c. ADDRESS (*Include Zip Code*)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME US ARMY EDGEWOOD RESEARCH, DEVELOPMENT AND ENGINEERING CENTER	b. TELEPHONE *Include Area Code)* (1) Commercial (410) 671-1879 (2) AUTOVON 584-1879
c. ADDRESS *(Include Zip Code)* ATTN: SCBRD-ENE-S ABERDEEN PROVING GROUND, MD 21010-5423	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: DEFENSE QUALITY AND STANDARDIZATION OFFICE 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22401-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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